QC810 - Task 5: Barren plateau risk analysis for quantum RL settings and survey quantum circuit ansätze.

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1 Introduction

Quantum Implicit Q-Learning (Q-IQL) aims to transpose the offline reinforcement learning framework of Kostrikov et al. into the parameterised quantum circuit setting, exploiting potential quantum advantages in function approximation while preserving the offline, safe training regime. The algorithm inherits the statistical and optimisation challenges of both its parents: the sparse, indirect gradient signal of offline RL, and the geometry-induced trainability pitfalls of VQAs.

Among these pitfalls, *barren plateaus* stand out as the most operationally dangerous. First identified analytically by McClean et al. [1], barren plateaus occur when the gradient of a VQA cost function concentrates exponentially around zero — meaning that an exponentially precise measurement is needed merely to determine a descent direction. Subsequent work by Cerezo et al. [3], Holmes et al. [4], and Grant et al. [2] has substantially refined our understanding of when, how severely, and how to mitigate this phenomenon.

The present report addresses the following concrete goals, set out in Issue #3 (parallel to Issue #4 on architecture mapping):

1. Review the barren plateau literature: origin, scaling behaviour, and known mitigation strategies.
2. Analyse the BP risk specifically in the Q-IQL setting: sparse rewards, offline data, expectile regression gradients.
3. Survey candidate ansätze for $V(s)$ approximation: hardware-efficient, problem-inspired, and data-reuploading circuits.
4. Evaluate ansätze along key axes: expressibility, entanglement capability, trainability, and circuit depth.
5. Review identity-block initialisation and local observables as mitigation strategies.
6. Provide concrete ansatz recommendations informed by the hybrid architecture of Issue #4: a quantum $V(s)$ network (4–8 qubits, angle encoding, data reuploading) paired with a classical $Q(s, a)$ network.

2 Barren Plateaus: Theory and Origins

2.1 Setup and Notation

Let $U(\boldsymbol{\theta}) : \mathcal{H} \to \mathcal{H}$ be a parameterised quantum circuit acting on n qubits, $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, $d = 2^n$. The cost function takes the standard VQA form

$$C(\boldsymbol{\theta}) = \mathrm{Tr}\big[H\, U(\boldsymbol{\theta})\, \rho\, U(\boldsymbol{\theta})^\dagger\big], \tag{1}$$

where ρ is an input state and H is a Hermitian observable. Training proceeds by minimising C with gradient-based classical optimisation; the partial derivative with respect to parameter θ_k is

$$\partial_k C = i\, \langle 0|\, U_-^\dagger \big[V_k,\, U_+^\dagger H U_+\big] U_-\, |0\rangle\,, \tag{2}$$

where $U_\pm$ are the circuit fragments above and below the k-th gate, and V_k is the corresponding generator [1].

2.2 The McClean 2018 Result: Exponential Gradient Vanishing

McClean et al. [1] showed that for random parameterised quantum circuits (RPQCs) — in particular those whose sub-circuits U_- or U_+ form a unitary 2-design — the expected gradient vanishes identically:

$$\mathbb{E}_U[\partial_k C] = 0, \tag{3}$$

and its variance decays exponentially with the number of qubits:

$$\mathrm{Var}[\partial_k C] \sim \begin{cases} -\dfrac{\mathrm{Tr}\big(\rho^2\big)}{2^{2n}}\ \mathrm{Tr}\big\langle [V_k, U_+^\dagger H U_+]^2 \big\rangle_{U_+} & U_- \text{ is 2-design,} \\ -\dfrac{\mathrm{Tr}\big(H^2\big)}{2^{2n}}\ \mathrm{Tr}\big\langle [V_k, U_- \rho U_-^\dagger]^2 \big\rangle_{U_-} & U_+ \text{ is 2-design,} \\ 2\,\mathrm{Tr}\big(H^2\big)\,\mathrm{Tr}\big(\rho^2\big)\left(\dfrac{\mathrm{Tr}\big(V^2\big)}{2^{3n}} - \dfrac{\mathrm{Tr}(V)^2}{2^{4n}}\right) & \text{both.} \end{cases} \tag{4}$$

The geometric picture is that of *concentration of measure*: Haar-random unitaries map $|0\rangle$ uniformly over the hypersphere of dimension $d - 1 = 2^n - 1$; any smooth function of the state concentrates exponentially around its mean [1]. The key corollary is that circuits of even modest depth — $\mathcal{O}(n^{1/d})$ on a d-dimensional qubit array — can reach the 2-design regime and thereby enter a barren plateau.

Remark 2.1. *The gradient being exponentially small does not merely slow training; it requires an exponentially large number of measurements to even resolve the descent direction from finite-sample noise. The cost of estimating a single gradient component scales as $\mathcal{O}(1/\epsilon^\alpha)$, in sharp contrast to the $\mathcal{O}(\log 1/\epsilon)$ scaling available classically.*

2.3 Cost-Function-Dependent Barren Plateaus

A critical extension by Cerezo et al. [3] shows that the barren plateau phenomenon is *not* solely a consequence of circuit depth or expressibility: it is also strongly dependent on the **locality of the observable** H.

Assuming an *alternating layered ansatz* (ALA) composed of m-qubit blocks forming local 2-designs, Cerezo et al. prove two complementary theorems:

Theorem 2.2 (Global cost ⇒ barren plateau; Cerezo et al., 2021)**.** *If C is defined by a global observable (one acting non-trivially on all n qubits), then* $\mathrm{Var}[\partial_\nu C]$ *vanishes exponentially in n for* any *circuit depth L. That is, global observables induce barren plateaus even in shallow circuits.*

Theorem 2.3 (Local cost ⇒ polynomial vanishing; Cerezo et al., 2021)**.** *If C is defined by an m-local observable (acting non-trivially on at most m qubits) and the circuit depth $L \in \mathcal{O}(\log n)$, then*

$$\mathrm{Var}[\partial_\nu C] \in \Omega\left(\frac{1}{\mathrm{poly}(n)}\right). \tag{5}$$

That is, for shallow circuits with local cost functions, the gradient vanishes at most polynomially.

An important ancillary phenomenon is the *narrow gorge*: even when the global cost avoids exact barren plateaus near its minimum, the probability of randomly landing in the region where $C \leq \delta$ decays exponentially with n (Proposition 1 in [3]). This means randomly initialised parameters are exponentially unlikely to be near the optimum, amplifying effective trainability difficulties even beyond the vanishing gradient.

2.4 Expressibility–Trainability Trade-off

Holmes et al. [4] complete the theoretical picture by showing that the barren plateau phenomenon extends continuously from the 2-design case to *arbitrary* ansätze, with the gradient variance bounded above by the ansatz **expressibility**:

Theorem 2.4 (Holmes et al., 2022)**.** *Let $\epsilon_{\mathbb{U}}^\rho$ and $\epsilon_{\mathbb{U}}^H$ measure the distance of the ansatz distribution from a 2-design (for the input state and observable, respectively). Then for any layered ansatz of the form of eq.* (1)*,*

$$\mathrm{Var}[\partial_k C] \leq \mathrm{Var}_R[\partial_k C] + 4\epsilon_R^\rho \, \|H\|_2^2, \tag{6}$$

$$\mathrm{Var}[\partial_k C] \leq \mathrm{Var}_L[\partial_k C] + 4\epsilon_L^H \, \|\rho\|_2^2, \tag{7}$$

where $\mathrm{Var}_{R/L}[\partial_k C]$ *are the variances evaluated when $U_\pm$ form exact 2-designs.*

The practical implication is stark: highly expressive ansätze (those that approximate 2-designs well) will have small ϵ, tight bounds, and correspondingly small gradient variance. Conversely, *low*-expressibility ansätze may support larger gradients — but only if they happen to contain the solution. There is thus a fundamental tension: expressibility is desirable for coverage of the solution space, but it directly drives gradient vanishing.

2.5 Additional Sources of Barren Plateaus

Beyond the sources analysed above, three further BP mechanisms are relevant to Q-IQL:

Noise-induced BPs. Wang et al. (2020) showed that hardware noise alone can induce BPs, independent of circuit depth or cost locality. For L layers and local depolarising noise rate λ, the effective signal decays as $(1-\lambda)^L$. This is particularly relevant in the NISQ regime where Q-IQL would be deployed.

Entanglement-induced BPs. Ortiz Marrero et al. (2020) proved that if the input state has volume-law entanglement, local cost values concentrate exponentially. This is relevant when encoding high-dimensional state representations.

Gradient-free optimisers. Arrasmith et al. (2020) demonstrated that the barren plateau problem equally affects gradient-*free* optimisers (Nelder-Mead, COBYLA, etc.). There is no escape through derivative-free methods; the loss landscape is exponentially flat regardless of the optimisation oracle.

3 Barren Plateau Risk in the Q-IQL Setting

Q-IQL inherits the variational quantum circuit optimisation framework and applies it to an offline reinforcement learning objective. We identify three compound risk factors that make BP particularly dangerous in this setting.

3.1 Risk Factor 1: Offline Data and Distributional Shift

In online RL, gradient signals from environment interaction provide a continual correction mechanism. In offline IQL, the agent optimises solely over a fixed replay buffer $\mathcal{D}$. This means:

- There is no ability to collect targeted data when gradients vanish.
- Small but persistent gradient errors from a barren plateau can, when accumulated over batches, produce systematic distributional shift in the learned value functions.
- Exponential overhead in gradient estimation is not amortisable over fresh experience.

3.2 Risk Factor 2: Sparse Rewards and Gradient Signal Dilution

Sparse reward environments produce cost functions C whose non-zero signal is concentrated in a small fraction of the training data. Concretely, if only a fraction f of transitions in $\mathcal{D}$ have non-zero reward, the effective gradient is diluted by a factor of f relative to the dense-reward case. Combining this with an exponentially small VQA gradient creates a doubly-exponential effective learning signal.

3.3 Risk Factor 3: Expectile Regression Gradients

Classical IQL trains the value function V_ψ via **expectile regression** at level $\tau \in (0,1)$:

$$\mathcal{L}_V(\psi) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}[\ell_\tau(Q_\phi(s,a) - V_\psi(s))]\,, \tag{8}$$

where $\ell_\tau(u) = |(\tau - \mathbf{1}[u<0])| \cdot u^2$.

For Q-IQL, this loss is implemented as the expectation value of a quantum observable. The expectile loss introduces two additional trainability risks:

a) **Asymmetric gradient weighting.** The asymmetry factor $|\tau - \mathbf{1}[u < 0]|$ reweights gradients by a data-dependent scalar, which can further suppress or amplify the already-exponentially-small VQA signal.

b) **Non-smooth loss.** The indicator function $\mathbf{1}[u < 0]$ introduces a non-smoothness that complicates gradient flow, and may interact poorly with the parameter-shift-rule-based gradient evaluation standard in VQAs.

3.4 Observable Design Implications

Combining the results of section 2.3 with the Q-IQL risk factors, we derive the following constraints on the choice of measurement observable:

I. **Global observables are forbidden.** By theorem 2.2, any $V(s)$ or $Q(s, a)$ network using a global observable (e.g., $\bigotimes_{i=1}^{n} \sigma_z^{(i)}$) will suffer exponential gradient vanishing regardless of ansatz choice or circuit depth.

II. **Local observables are necessary (though not sufficient).** By theorem 2.3, k-local observables with $L \in \mathcal{O}(\log n)$ guarantee at most polynomial gradient vanishing. For Q-IQL, this means measuring only a small subset of qubits (e.g., a single qubit readout, or a sum of single-qubit Pauli terms).

III. **Locality must be maintained through training.** Growing the circuit adaptively (layer-by-layer training) must preserve observable locality at each stage.

3.5 Risk Scope: Quantum $V(s)$ Only

A crucial observation from the Issue #4 architecture is that **the $Q(s, a)$ network is entirely classical**. This means barren plateau risk is confined exclusively to the quantum $V(s)$ component. The classical $Q(s, a)$ network faces the standard challenges of deep learning (vanishing gradients, overfitting) but is immune to the exponential-in-n gradient vanishing characteristic of VQAs. All BP analysis and mitigations in the remainder of this report apply solely to $V(s)$.

3.6 Compound Risk Summary

Table 1: Compound barren plateau risk factors in the Q-IQL quantum $V(s)$ network.

Risk Factor	**Mechanism**	**Severity**
Deep/expressive circuits	$\mathrm{Var}[\partial C] \sim 2^{-n}$	Critical
Global observables	BP at any depth (Cerezo [3])	Critical
Offline data	No corrective data collection	High
Sparse rewards	Gradient dilution by reward density f	High
Expectile asymmetry	Asymmetric gradient suppression	Medium
Hardware noise	Noise-induced BPs $(1 - \lambda)^L$ decay	Medium–High

4 Mitigation Strategies

4.1 Identity-Block Initialisation (Grant et al., 2019)

Grant et al. [2] propose a practically effective initialisation strategy that addresses the BP at the start of training. The key idea is to initialise the circuit as a sequence of M *shallow blocks*, each of which evaluates to the identity:

$$U_m(\boldsymbol{\theta}_m^{\text{init}}) = \left(\prod_{l=1}^{L} U_l(\theta_{l,1}^m)\right)\left(\prod_{l=L}^{1} U_l(\theta_{l,1}^m)^\dagger\right) = \mathbf{1}, \tag{9}$$

so that the full circuit $U(\boldsymbol{\theta}^{\text{init}}) = \prod_m U_m = \mathbf{1}$.

When a single parameter θ_k in block m is updated, the effective circuit depth seen by the gradient computation is the *block depth* L, not the full depth ML. Provided L is small enough that individual blocks do not approach a 2-design, this guarantees non-exponentially-small gradients at the first training step.

Empirical validation. Grant et al. demonstrated that:

- For random initialisation with 120-layer circuits, the gradient variance decays exponentially with n as predicted.
- For identity-block initialisation (single block, $L = 60$ layers), the variance is *invariant* to n up to $n = 12$ qubits tested.
- QNN classifiers trained with identity-block initialisation never encountered the barren plateau during training, with gradient variance decreasing only as the model converged.

Caveat for VQE/Q-IQL. The identity initialisation does not generate entanglement at the outset. This is resolved by prepending a fixed shallow entangling layer B (not updated during training) that introduces initial correlations without contributing to barren plateau depth.

4.2 Local Cost Functions

As established in section 2.3, restricting the observable to act locally is the theoretically strongest mitigation. For Q-IQL, this translates to:

- $V(s)$**:** Readout from a single designated qubit via $\sigma_z^{(j)}$ or a sum $\frac{1}{n}\sum_j \sigma_z^{(j)}$.
- $Q(s,a)$**:** Similarly, a local readout with the action a encoded in the input rather than the measurement.

Note that local observables have a subtler *faithfulness* requirement: the local cost must vanish under the same conditions as the corresponding global cost. Cerezo et al. proved that their proposed local autoencoder cost C_L satisfies $C_L \leq C_G \leq n_B C_L$, ensuring faithfulness. An analogous faithful local cost must be constructed for the Q-IQL value objectives.

4.3 Layer-wise Training

Skolik et al. (2020) proposed incrementally growing the circuit depth during training: begin with a shallow ansatz (provably trainable by theorem 2.3), and append layers as earlier layers converge. This is particularly natural for Q-IQL, where the curriculum of offline data can be structured to match the growing circuit expressibility.

4.4 Parameter Correlation and Restricted Rotation Angles

Holmes et al. [4] numerically demonstrated that two additional strategies significantly amplify cost gradients:

a) **Correlating parameters across qubits and layers.** When all qubits rotate by the same angle in the same direction, the ansatz is rendered highly inexpressive, yielding near-constant (non-vanishing) gradient variance. The trade-off is that the solution may not lie within the reduced parameterisation.

b) **Initialising near the identity with restricted angle range.** When $\theta_{l,i}^{\text{init}} = 0$ and angles are restricted to $[\tilde{\theta}, \tilde{\theta} + 2\pi r]$ with small r, gradient vanishing is suppressed for local costs. This is essentially a continuous version of the identity-block strategy.

5 Survey of Candidate Ansätze

We survey three families of ansätze relevant to Q-IQL function approximation, evaluating each along four axes:

- **Expressibility (Expr):** Coverage of the unitary group; distance from 2-design.
- **Entanglement capability (Ent):** Ability to generate entangled output states.
- **Trainability (Train):** Resistance to barren plateaus; gradient scaling.
- **Circuit depth (Depth):** Gate count; relevance for NISQ hardware.

5.1 Hardware-Efficient Ansatz (HEA)

Structure. The HEA [6] alternates layers of single-qubit rotations with fixed entangling layers (typically CNOT or CZ ladders). For L layers on n qubits, the circuit depth is $\mathcal{O}(Ln)$ with $\mathcal{O}(Ln)$ parameters.

Expressibility. High: with sufficient depth, HEA circuits approximate 2-designs. This is precisely the regime that McClean et al. [1] show leads to BPs.

Entanglement capability. High: the nearest-neighbour CZ structure creates extensive entanglement. However, Ortiz Marrero et al. showed that high entanglement in the output state itself constitutes an additional source of BPs.

Trainability. Poor for deep circuits. As shown by [1, 3, 4], deep HEA with global observables exhibits exponential gradient vanishing even for shallow ($L = \mathcal{O}(1)$) circuits when global observables are used. With local observables and $L \in \mathcal{O}(\log n)$, gradient vanishing is at worst polynomial.

Circuit depth. Low-to-moderate. The hardware-native gate set minimises overhead, making HEA the most NISQ-friendly option for small qubit counts.

Assessment for Q-IQL. HEA is viable for small system sizes ($n \lesssim 10$) when combined with: (a) identity-block initialisation, (b) local readout observables, and (c) $L \in \mathcal{O}(\log n)$ depth. Without all three mitigations simultaneously, BP risk is high.

5.2 Problem-Inspired / Hamiltonian Variational Ansatz

Structure. Problem-inspired ansätze (PIAs) encode domain knowledge in the circuit structure. The Hamiltonian Variational Ansatz (HVA) uses alternating layers of operations generated by the Hamiltonian terms [7]. For Q-IQL, a natural PIA would encode the structure of the MDP transition operator.

Expressibility. Low-to-moderate: PIAs access a restricted subset of the unitary group, determined by the Hamiltonian structure. By theorem 2.4, lower expressibility implies larger upper bounds on gradient variance.

Entanglement capability. Problem-dependent: HVA generates entanglement patterns dictated by the Hamiltonian connectivity.

Trainability. Good, precisely because low expressibility keeps the ansatz away from 2-designs. Holmes et al. [4] note that problem-inspired ansätze are the recommended approach for avoiding BPs while maintaining solution coverage.

Circuit depth. Moderate-to-high: faithfully encoding the MDP structure may require deep circuits, depending on the state and action space dimensionality.

Assessment for Q-IQL. Given that Issue #4 specifies angle encoding and data reuploading as the encoding strategy for $V(s)$, PIA is not the primary choice for this project. It remains a theoretically sound alternative if future work identifies exploitable structure in the MDP transition dynamics.

5.3 Data Reuploading Circuits (DRU)

Structure. Pérez-Salinas et al. [5] showed that a single qubit can serve as a universal classifier when classical data is re-uploaded at each layer:

$$U(\boldsymbol{\phi}, \mathbf{x}) = U(\boldsymbol{\phi}_N)U(\mathbf{x}) \cdots U(\boldsymbol{\phi}_1)U(\mathbf{x}), \tag{10}$$

where each layer $L(i) = U(\boldsymbol{\phi}_i)U(\mathbf{x})$ combines a data-encoding rotation with trainable parameters. More compactly, data and parameters can be combined: $L(i) = U(\boldsymbol{\theta}_i + \mathbf{w}_i \circ \mathbf{x})$.

The circuit can be understood as a quantum analogue of a neural network with a single hidden layer, where re-uploading bypasses the no-cloning theorem: classical data is re-introduced at each layer without quantum copying. The universality result follows from an analogy with the Universal Approximation Theorem: the trigonometric functions of the layer rotations form a Fourier-like basis sufficient to approximate any continuous function [5].

Extension to multiple qubits. The DRU architecture extends naturally to k qubits with entanglement (e.g., CZ gates after each reuploading layer). Pérez-Salinas et al. demonstrated that multi-qubit DRU with entanglement achieves higher accuracy with fewer layers than the single-qubit version across a range of benchmarks.

Expressibility. Moderate: DRU circuits with many layers achieve high expressibility, but the re-uploading structure constrains the accessible unitary space relative to unconstrained HEA of the same depth.

Entanglement capability. Configurable: entanglement is introduced via explicit inter-qubit gates between reuploading layers. Without these gates, each qubit is an independent single-qubit classifier.

Trainability. Promising. The alternating data-encoding structure breaks the homogeneity that drives 2-design convergence. Each layer's rotation angle is constrained to $\boldsymbol{\theta}_i + \mathbf{w}_i \circ \mathbf{x}$, which limits how deeply the circuit can scramble the input. However, with many layers and full entanglement, DRU circuits can still approach 2-designs.

Circuit depth. Moderate: DRU circuits of N layers have depth $2N$ (data + processing per layer), or N in the compressed form. For single-qubit DRU, Pérez-Salinas et al. achieved $> 90\%$ accuracy with $N = 2$ layers (12 parameters) on several benchmarks.

Assessment for Q-IQL. DRU is particularly attractive for Q-IQL because:

- State observations s and actions a are naturally re-uploaded at each layer, encoding richer input–circuit interactions than a single-layer encoding.
- The fidelity cost function and weighted fidelity cost function analysed by Pérez-Salinas et al. can serve as local cost proxies for $V(s)$ and $Q(s, a)$.
- The single-qubit readout is a natural local observable, directly satisfying the locality constraint from section 3.4.

The main open question is whether the re-uploading structure alone suffices to suppress BPs, or whether additional identity-block or layerwise training mitigations are needed.

Table 2: Comparative evaluation of candidate ansätze for the quantum $V(s)$ network in Q-IQL. Ratings: **H**igh / **M**edium / **L**ow. BP risk assumes local observables and identity-block initialisation unless noted. The $Q(s, a)$ network is classical and is not evaluated here.

Ansatz	**Expr.**	**Entangl.**	**Train.**	**Depth**	**Notes**
HEA (shallow, local obs.)	M	M–H	M	L	Viable with mitigations; BP risk grows with depth
HEA (deep, global obs.)	H	H	L	H	High BP risk; not recommended
PIA / HVA	L–M	Problem-dep.	H	M–H	Preferred if MDP structure known; not primary choice
DRU (single qubit)	M	L	M–H	M	Natural local readout; lower entanglement
DRU (multi-qubit, entangled)	**M–H**	**M**	**M–H**	**M**	**Recommended for $V(s)$; best balance**

6 Comparative Table and Recommendations

6.1 Comparative Ansatz Table

6.2 Architecture Context from Issue #4

The hybrid forward pass proposed in Issue #4 has the following structure:

- **Quantum branch ($V(s)$):** A classical encoder f_{enc} maps state s to a feature vector x_s, which is fed into a parameterised quantum circuit $U(\boldsymbol{\theta}, x_s)$ using angle encoding with data reuploading. A local observable O_V is measured to produce $\hat{V}(s)$. Circuit size: 4–8 qubits.
- **Classical branch ($Q(s, a)$):** A separate classical encoder g_{enc} maps the joint (s, a) pair to features x_{sa}, which are processed by a standard neural network to produce $Q(s, a)$.

This design has two important consequences for BP analysis. First, the classical $Q(s, a)$ network carries zero BP risk by construction. Second, the angle encoding + data reuploading specification for $V(s)$ directly aligns with the DRU ansatz family surveyed in section 5.3, making the ansatz choice straightforward.

6.3 Recommendation for $V(s)$

Recommended ansatz: multi-qubit data reuploading circuit with CZ entanglement and local single-qubit readout.

The concrete circuit for $V(s)$ should be structured as follows:

Encoding. The pre-encoded feature vector $x_s \in \mathbb{R}^d$ (output of f_{enc}) is angle-encoded into the qubit rotations at each layer. Each reuploading layer takes the form

$$L^{(i)} = \text{Ent} \cdot \bigotimes_{j=1}^{n} R_j\left(\boldsymbol{\theta}_j^{(i)} + \mathbf{w}_j^{(i)} \circ x_s\right), \tag{11}$$

where $R_j(\cdot)$ is a general single-qubit rotation on qubit j, $\boldsymbol{\theta}_j^{(i)}$ are trainable bias parameters, $\mathbf{w}_j^{(i)}$ are trainable data weights, and Ent is a fixed layer of nearest-neighbour CZ gates. The full circuit is $U(\boldsymbol{\theta}, x_s) = L^{(N)} \cdots L^{(1)}$.

Qubit count. For 4–8 qubits as specified, the gradient variance under local cost functions scales polynomially (by theorem 2.3), making this range safe from exponential BP provided $L \in \mathcal{O}(\log n)$. For $n = 8$, this means $L \lesssim 3$ layers before the polynomial guarantee degrades. Deeper circuits remain viable but require empirical gradient monitoring.

Observable. The readout observable O_V must be local. We recommend a single-qubit Pauli-Z measurement:

$$\hat{V}(s) = \langle \sigma_z^{(1)} \rangle = \langle 0 | \, U(\boldsymbol{\theta}, x_s)^\dagger \, \sigma_z^{(1)} \, U(\boldsymbol{\theta}, x_s) \, | 0 \rangle \, . \tag{12}$$

Alternatively, an average over a small subset of qubits, $O_V = \frac{1}{k} \sum_{j=1}^{k} \sigma_z^{(j)}$ with $k \ll n$, preserves locality while providing a smoother readout signal.

Initialisation. Apply identity-block initialisation: set $\boldsymbol{\theta}_j^{(i)}$ randomly and $\mathbf{w}_j^{(i)} = -\boldsymbol{\theta}_j^{(i)}$ so that each layer evaluates to the identity at the start of training. Prepend a fixed, non-trained shallow entangling layer to introduce initial correlations.

6.4 On $Q(s, a)$: No BP Considerations

Since $Q(s, a)$ is implemented as a classical neural network in the Issue #4 architecture, no barren plateau analysis applies. Standard deep learning practices (Xavier/He initialisation, batch normalisation, Adam optimiser) are sufficient. The interaction between the classical Q-network and the quantum V-network occurs only through the shared IQL training objective — specifically through the expectile regression loss, which uses $Q(s, a)$ as a target for $V(s)$. The BP risk in this interaction path is one-directional: noisy or slow-converging $V(s)$ gradients (due to BP) can affect the Q-network's target quality, but not vice versa.

6.5 Mitigation Checklist for $V(s)$

The following mitigation stack is recommended for the quantum $V(s)$ network:

M1. Local observable. Use $O_V = \sigma_z^{(1)}$ or a k-local sum with $k \leq 2$. Never use a global observable such as $\bigotimes_j \sigma_z^{(j)}$.

M2. Identity-block initialisation. Set initial parameters so that $U(\boldsymbol{\theta}^{\text{init}}, x_s) = \mathbf{1}$ for any x_s. For DRU, this means $\mathbf{w}^{(i)} = -\boldsymbol{\theta}^{(i)}$ at initialisation.

M3. Fixed entangling preamble. Prepend a fixed shallow CZ layer to generate initial entanglement without contributing to BP depth.

M4. Shallow circuit. Keep $L \in \mathcal{O}(\log n)$ layers to maintain polynomial gradient guarantees. Monitor gradient variance empirically as L grows.

M5. Layer-wise training. Train with $L = 1$ layers first, then progressively add layers as the shallower circuit converges.

M6. Faithful local cost. Verify that $\langle \sigma_z^{(1)} \rangle = 0$ at the optimum of the expectile loss, i.e., that the local observable faithfully represents the global training objective.

7 Remaining Open Questions

With the architecture of Issue #4 now defined, most of the structural unknowns have been resolved. The following questions remain open for future experimental work:

1. **Optimal layer count L for 4–8 qubits.** The theoretical guarantee of polynomial gradient vanishing holds for $L \in \mathcal{O}(\log n)$. For $n = 4$ this allows $L \leq 2$; for $n = 8$, $L \leq 3$. Whether deeper circuits remain practically trainable should be verified empirically by monitoring $\text{Var}[\partial_k C]$ during early training.

2. **Optimal data weight initialisation for DRU.** The identity-block strategy for DRU requires $\mathbf{w}^{(i)} = -\boldsymbol{\theta}^{(i)}$, meaning data weights cancel the bias at initialisation. The rate at which data weights should be allowed to grow during training (i.e., the angle range schedule) requires empirical tuning.

3. **Expectile loss faithfulness with local observable.** A formal proof that $\langle \sigma_z^{(1)} \rangle = 0$ iff the expectile regression loss is minimised remains to be established, analogous to Cerezo et al.'s Proposition 3 for the autoencoder case.

4. **Interaction between classical Q-network convergence and quantum V gradients.** In the IQL training loop, $Q(s, a)$ provides targets for $V(s)$. If the classical network converges faster, noisy $V(s)$ gradients may cause instability in Q-targets. The interplay between the two learning timescales warrants study.

5. **Hardware noise budget.** For NISQ deployment, the noise-induced BP threshold $(1 - \lambda)^L$ should be estimated for the target hardware to inform the maximum feasible circuit depth.

8 Conclusion

Barren plateaus represent a critical risk for the quantum $V(s)$ component of Q-IQL, compounded by three simultaneous factors: the intrinsic exponential gradient vanishing of VQAs, the indirect training signal of offline RL with sparse rewards, and the asymmetric gradient weighting introduced by expectile regression. The classical $Q(s, a)$ network in the Issue #4 architecture is exempt from BP by construction.

The hybrid architecture of Issue #4 — a quantum $V(s)$ network with 4–8 qubits, angle encoding, and data reuploading, paired with a classical $Q(s, a)$ network — is well-aligned with the theoretical prescriptions of the BP literature. The data reuploading structure inherently constrains expressibility relative to unconstrained HEA, the angle encoding provides a natural channel for state information at each layer, and the small qubit count ($n \leq 8$) keeps the gradient variance in the polynomially-vanishing regime under local observables for shallow circuits.

The recommended implementation for $V(s)$ combines multi-qubit DRU layers (eq. (11)) with a local single-qubit readout (eq. (12)), identity-block initialisation, and a fixed shallow entangling preamble. The six-point mitigation checklist in Section 6 provides actionable guidance for the implementation phase.

Key open questions — the optimal layer count for empirical trainability, the data weight initialisation schedule, and the formal faithfulness of the local expectile loss — remain for experimental validation.

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